Lecture 10 Supplement – `ascentopt`: Launch Trajectory & Propulsion Co-Design

Mechanics & Thermodynamics of Propulsion (MECE 4305)

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Why a Trajectory Optimizer

A forward simulation flies a prescribed law; design needs the law solved for

The usual first look at a launch is a 3-DOF forward simulation: vertical rise, then a gravity turn (thrust along velocity). One stage, prescribed steering, no constraints, no objective.

- it answers "where does this law go?" – not "what is the most this vehicle can do?";
- the loss rule of thumb, $\Delta v_{loss} \approx 1.5$–$2.0\,\mathrm{km/s}$, is a band, not a number;
- a design needs: payload for a *given* vehicle, the staging point, the loss breakdown, and the propulsion requirements T, I_{sp}, $\dot{m}$, burn time for the thrust-chamber and cycle steps.

What is needed: trajectory optimization with a multistage vehicle model – that is `ascentopt`.

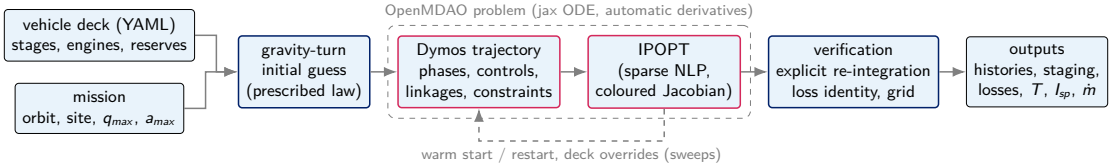
Hand method vs. the Starship Block 3 deck (dry 200/100 t)

assumed Δv_{req}	loss assumption	payload
8800 m/s	1.5 km/s	339 t
9300 m/s	2.0 km/s	276 t
9155 m/s	**optimizer:** 1833 m/s	293 t

The trajectory decides the loss; the loss decides the payload. The band is worth ± 30 t here.

What ascentopt Is

3-DOF ascent physics – steering and throttle become unknowns, the orbit a constraint, payload the objective



Physics (as in a forward simulation)

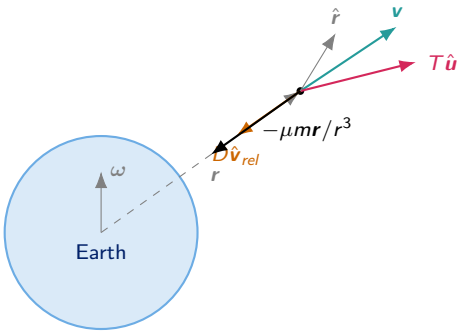
3-DOF point mass, rotating Earth, atmosphere and drag, back-pressure thrust law; a prescribed gravity-turn law supplies the initial guess and an independent check.

Added

Controls $\hat{\boldsymbol{u}}(t)$, $\tau(t)$; phases linked at staging; path and terminal constraints; max-payload objective; five loss accumulators; sweeps by deck override.

Physical Model

Point mass in an Earth-centred inertial frame; drag on the air-relative velocity



$$\dot{\boldsymbol{r}} = \boldsymbol{v}$$

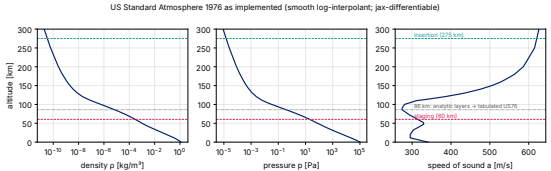
$$\dot{\boldsymbol{v}} = -\tfrac{\mu}{r^3}\boldsymbol{r} + \tfrac{T}{m}\hat{\boldsymbol{u}} - \tfrac{D}{m}\hat{\boldsymbol{v}}_{rel}, \qquad \boldsymbol{v}_{rel} = \boldsymbol{v} - \boldsymbol{\omega} \times \boldsymbol{r}$$

$$\dot{m}_p = -\tau\,\dot{m}_{max}, \qquad T = \tau\left(T_{vac} - p_a(h)A_e\right), \qquad D = \tfrac{1}{2}\rho(h)|\boldsymbol{v}_{rel}|^2 C$$

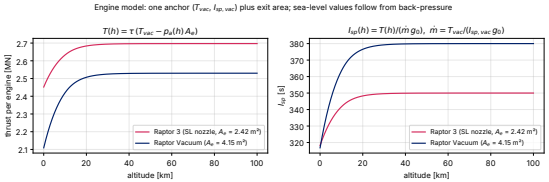
- ECI Cartesian: no polar singularity, no trigonometry in the dynamics
- $\boldsymbol{v}_0 = \boldsymbol{\omega} \times \boldsymbol{r}_0$: the 418 m/s rotation credit falls out – no separately signed term to get wrong
- drag on the *air-relative* velocity, not the inertial velocity
- loss rates $g\sin\gamma$, D/m, $(T/m)(1-\cos\alpha_T)$, $\tau p_a A_e/m$ integrated as extra states

Environment and Engine Models

US Standard Atmosphere 1976 as a smooth table; one anchor pair per engine



Analytic layers to 86 km, tabulated above, log-interpolated $\Rightarrow$ differentiable to orbit. Matches the `ambiance` ISA package to $< 0.1\,\%$ below 81 km.

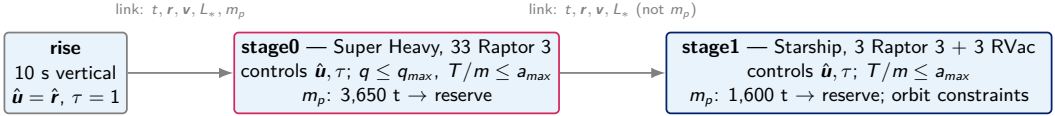


(T_{vac}, $I_{sp_{,vac}}$) + exit area; sea-level values are *outputs*. Published Raptor pairs from different sources are not mutually consistent (258 tf vac would need a 1.86 m exit) – a model, not a lookup.

The Optimal-Control Problem

Maximize payload; one phase per burn, propellant a separate state per stage

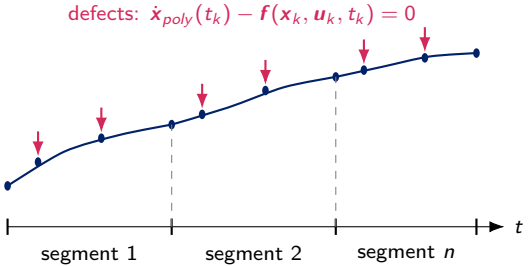
maximize	m_{pay} (or: minimize propellant, minimize time)
over	$\hat{\boldsymbol{u}}(t)$, $\tau(t)$ per burn phase; phase durations; m_{pay}
dynamics	collocation defects in every phase
path	$\lvert\boldsymbol{u}\rvert^2 = 1$, $0.4 \le \tau \le 1$, $h \ge 0$, $q \le 35\,\text{kPa}$, $T/m \le 4\,g_0$
linkage	$t, \boldsymbol{r}, \boldsymbol{v}$ and loss states continuous between phases; m_p is *not* linked
boundary	m_p = stage load at ignition; $m_p(t_f) \ge m_{reserve}$ at burnout
terminal	$h = 275\,\text{km}$, $\gamma = 0$, $\lvert\boldsymbol{v}\rvert = \sqrt{\mu/r}$ $[,\ i = i_{target}]$



Design procedure, step 0–4: mission → Δv → propellant fraction → required I_{sp} → staging, solved as one problem with the trajectory in the loop.

Direct Collocation in `ascentopt`

Radau pseudospectral transcription, exact sparse derivatives, IPOPT



Legendre–Gauss–Radau, order 3, 16 segments per phase.
Two-stage case: 1,288 variables, 1,474 constraints, 1.1 % dense, 43 colours.

Initial guess = prescribed gravity turn with a tuned pitch kick.
Cold start ≈260 IPOPT iterations; warm-started sweep cases 20–50.

Why it converged – three facts

1. **Exact, sparse derivatives.** ODE as an OpenMDAO `JaxExplicitComponent`; node-diagonal sparsity *declared* (undeclared: 504 colours; declared: 43).
2. **Smooth outputs.** γ, α, i via $\arctan 2$ – $\arcsin$ / $\arccos$ have infinite slope exactly at a gravity-turn guess (10^9 dual infeasibility). Insertion as $\gamma = 0$, not $\boldsymbol{r} \cdot \boldsymbol{v} = 0$ (10^{10} scale).
3. **IPOPT**, not SLSQP: SLSQP failed Dymos's own SSTO benchmark in 3,000 iterations; IPOPT solves it in 27 (143.77 s vs documented ≈143 s).

Example: Starship / Super Heavy Block 3 – Inputs

Published propellant loads and Raptor 3 performance; dry masses are not published and are swept

	Super Heavy	Starship
propellant	3650 t	1600 t
dry mass (baseline, *swept*)	200 t	100 t
engines	33 × Raptor 3	3 Raptor 3 + 3 RVac
stage T_{vac}, $\dot{m}$	89.0 MN, 25 930 kg/s	15.7 MN, 4394 kg/s
T/W	1.46 at liftoff (SL)	0.89 at ignition
throttle	0.40–1.00	0.40–1.00

Raptor 3: 250 tf SL / 275 tf vac, $I_{sp,vac}$ 350 s $\Rightarrow$ A_e 2.42 m^2, $I_{sp,SL}$ 318 s. RVac: 258 tf, 380 s, 2.3 m exit. 9 m diameter, C_D 0.5.

Mission

Starbase 26.0°N $\rightarrow$ 275 km circular (Flight 14's orbit), inclination free, $q \leq 35$ kPa, $T/m \leq 4\,g_0$, 10 s vertical rise.

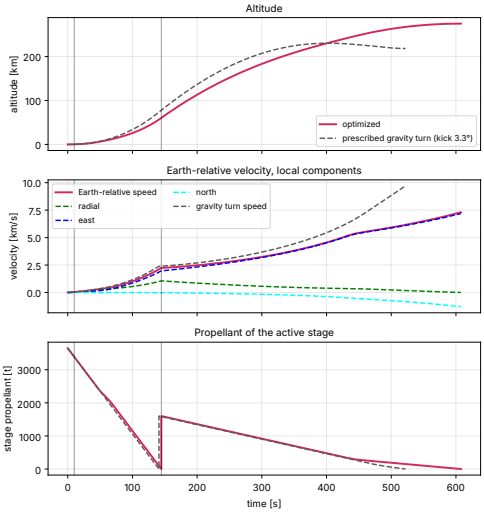
Unknowns treated as inputs

Dry masses are not published: estimate ranges 160–200 t / 85–100 t, mid-points as baseline. Return-propellant reserves are allowances, swept, not flown.

SpaceX figures as compiled on Wikipedia (Block 3 propellant loads; Raptor 3 May 2026 / Aug 2024; RVac 258 tf, 380 s). RVac exit diameter assumed; immaterial above staging altitude.

Baseline Solution – Trajectory

Red: optimized. Grey dashed: a prescribed gravity-turn law applied to the same vehicle



- the gravity turn (kick 3.3°) lofts to 218 km with 10.1 km/s – 2.4 km/s more than circular;
- the optimizer flies lower and flatter, inserts at exactly 275 km / 7740 m/s, and carries the difference as payload;
- both stages burn to zero reserve; staging at 145 s.

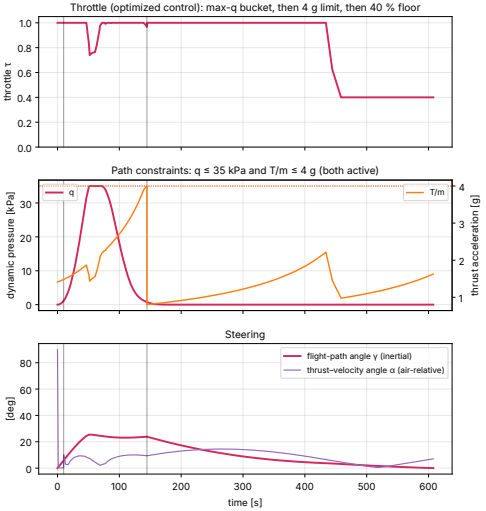
293.3 t

payload to 275 km × 26.0°, expendable

trajectory, dry 200/100 t

Baseline Solution – Controls and Active Constraints

Both path constraints are active: max-q in the booster burn, 4 g_0 at MECO



- **max-q bucket**: throttle to 0.74 between 50 and 75 s holds $q = 35$ kPa exactly;
- **acceleration limit**: the booster reaches 4 g_0 at MECO; the ship throttles to its 40 % floor for the last 170 s;
- **steering**: thrust pulls 10–15° off the air-relative velocity during the ship burn (the 690 m/s steering loss) while γ decays smoothly to zero.

Baseline Solution – the Numbers

Losses total 1833 m/s: inside the textbook 1.5–2.0 km/s band, now decomposed

293.3 t

payload to 275 km × 26.0°

(expendable trajectory, dry 200/100 t)

145 s · 60 km · 2.61 km/s

staging (MECO), at the 4 g_0 limit

608 s

insertion: h 275.000 km, γ 5×10^{-10} deg, 7740.26 m/s

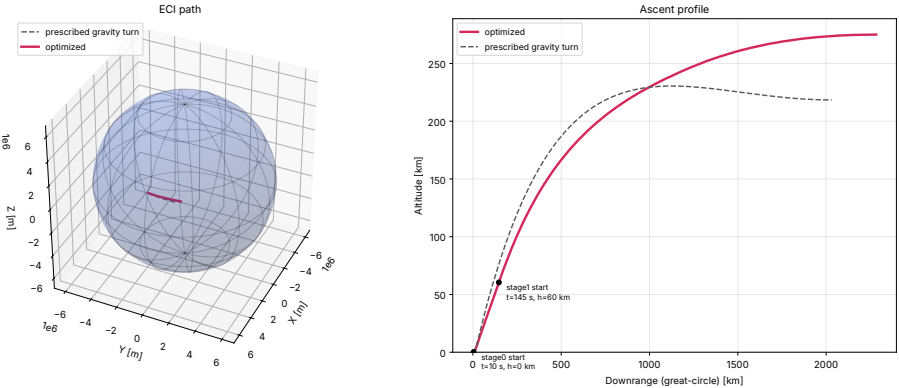
Trajectory		*Δv budget (m/s)*	
GLOW	5843 t	Δv_{vac} (both stages)	9,155
max-q (active)	35.0 kPa, 50–75 s	speed gained $\lvert \boldsymbol{v}_f \rvert - \lvert \boldsymbol{v}_0 \rvert$	7,322
booster min throttle	0.74	gravity loss	1,049
max T/m (active)	4.00 g_0 at MECO	steering loss	690
ship burn	463 s; $\tau = 0.40$ last 170 s	back-pressure loss	76
inclination	25.997° (= latitude)	drag loss	18
both stages	burn to zero reserve	$\lvert \boldsymbol{v}_0 \rvert$ (rotation)	418
		identity residual	−0.001

Design procedure, step 3: the losses are now computed for this vehicle and this orbit rather than taken from a band; the identity $\Delta v_{ideal} = v_{orbit} + \sum \text{losses} - v_{rot}$ closes to 1 mm/s.

The Same Vehicle Under a Prescribed Gravity Turn

Simulation vs. optimization: globe view and ascent profile

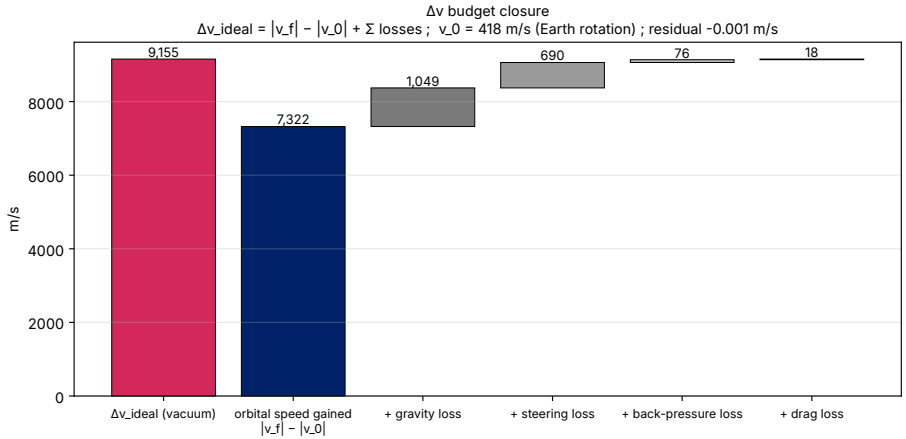
Starship Block 3 → 275 km × 26° (Starbase), expendable trajectory, dry 200/100 t



The prescribed law reaches $\gamma = 0$ at 218 km with 10.1 km/s: an eccentric orbit with 2.4 km/s to spare. The optimizer inserts into the 275 km circle and carries the difference as 293 t of payload.

The Δv Budget, Closed Numerically

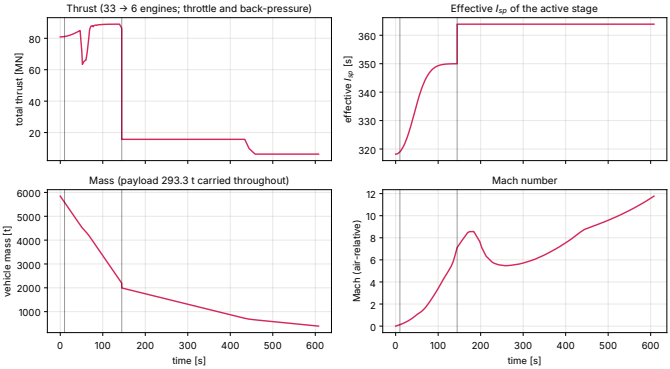
$\Delta v_{ideal} = v_{orbit} + \sum \text{losses} - v_{rot}$ – computed instead of assumed



$\Delta v_{vac} = (|\boldsymbol{v}_f| - |\boldsymbol{v}_0|) + L_g + L_s + L_{bp} + L_D$ holds to $-0.001\,\text{m/s}$ because the five terms are integrated as states alongside the dynamics.

Propulsion Requirements Handed to the Thrust-Chamber and Cycle Steps

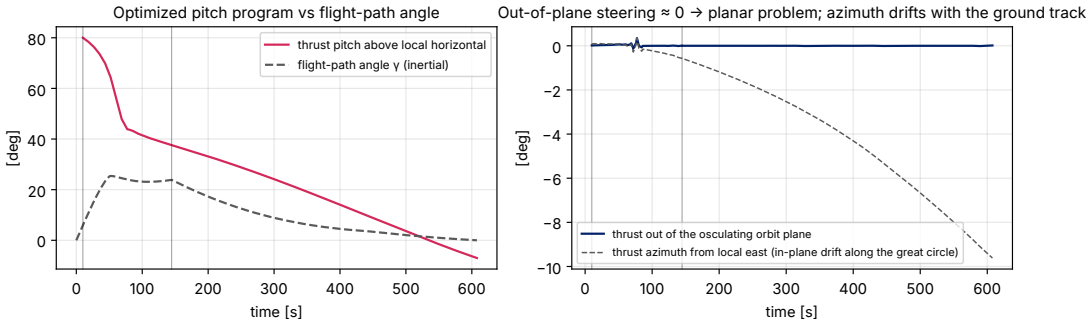
Per stage: T_{vac}, throttle profile, $I_{sp}(h)$, $\dot{m}$, burn time, propellant



Design procedure, step 5–8: these are the inputs to thrust-chamber design and to cycle / turbopump design; the I_{sp} rise in the first 60 s is the back-pressure term, the step at 145 s is staging onto the Raptor 3 / RVac set (thrust-weighted 363.9 s).

What the Optimizer Chose: Steering

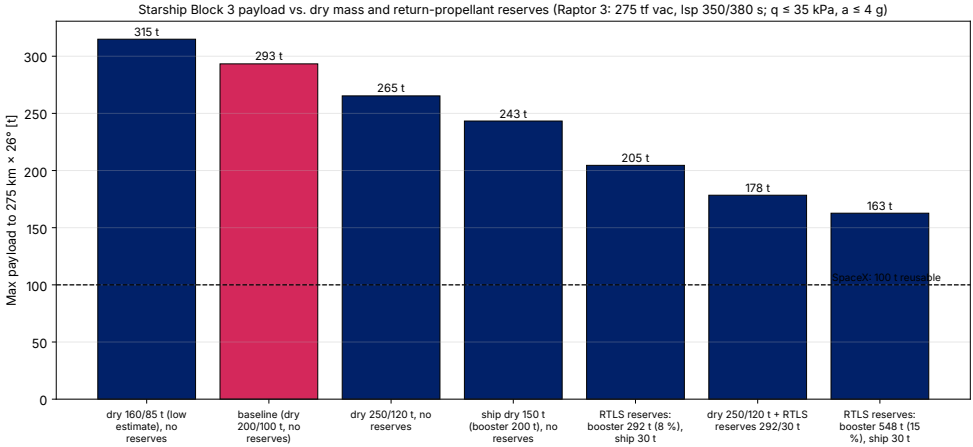
Pitch-over through the max-q bucket, then a near-linear pitch program; no out-of-plane steering



Pitch to $\sim 45^\circ$ by the end of the max-q bucket, then down to slightly negative at insertion. The thrust component out of the osculating orbit plane is zero: the optimizer discovers that an eastward launch is a planar problem; the azimuth drifts only because the ground track leaves the 26°N apex of the orbit.

What the Published 100 t Depends On

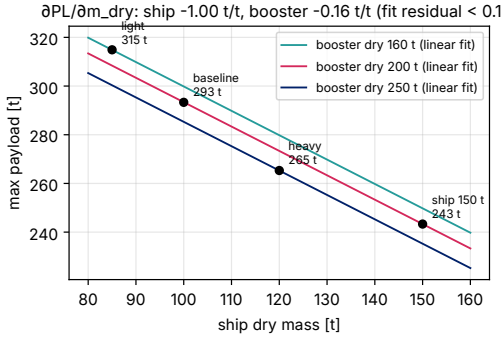
Seven cases from the baseline by command-line deck overrides, each warm-started from the previous solution

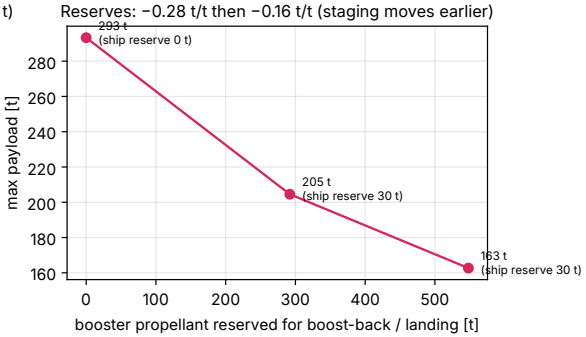


Sensitivities

Ship dry mass 1:1, booster dry mass 0.16:1, reserves the largest lever

Sensitivities extracted from the sweep (dry 200/100 t baseline, 275 km × 26°)



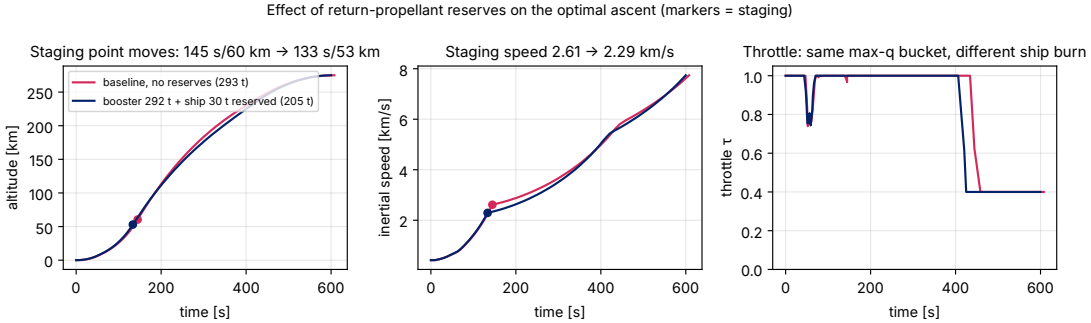


$m_{pay} = 425.8 - 0.161\, m_{dry,booster} - 1.001\, m_{dry,ship}$ [t]
reproduces the four no-reserve cases to 0.07 t: a kilogram on the ship is a kilogram of payload; on the booster, 160 g.

8 % of booster propellant held back + 30 t in the ship costs 89 t; 15 % costs 131 t. The optimizer re-stages earlier on its own – the mechanism behind "stage low for reusability".

Reserves Move the Staging Point

Same max-q bucket; MECO moves earlier and lower and the ship carries more of the Δv



MECO: 145 s / 60 km / 2.61 km/s (no reserves) → 133 s / 53 km / 2.29 km/s (8 %) → 123 s / 47 km / 2.00 km/s (15 %).

IFT-3 public timeline: hot staging at T+2:50, 72 km, on a flown return-to-site profile (Cité de l'espace)

Where the Remaining Gap to 100 t Lives

The pessimistic corner (dry 250/120 t + 8 % reserve) still gives 178 t

What the model carries

- published propellant loads and Raptor 3 performance
- dry masses (swept 160–250 / 85–150 t)
- return-propellant allowances (0–15 % booster, 30 t ship)
- max-q and acceleration limits, throttle floor
- gravity, drag, steering and back-pressure losses – computed, not assumed

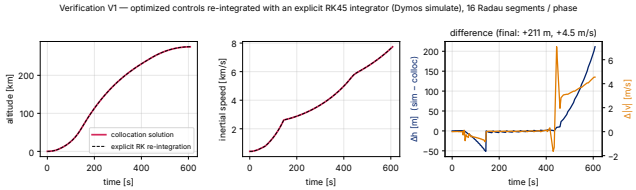
What it does not carry (yet)

- actual ship dry mass with heat shield and flaps (likely > 120 t) – 1:1 on payload
- boost-back propellant that grows with staging speed and must fly a *specific* return trajectory
- engine margins, $C_D(M)$, J_2, winds, 6-DOF
- operational and performance margins

The tool's contribution is not the number – it is that the gap is now a list of named quantities with known sensitivities.

Verification

What was actually checked (9 Nov 2026)



check	result
Dymos SSTO benchmark	t_f 143.77 s, 27 iter. (doc ≈143 s)
explicit re-integration	+211 m, +4.5 m/s (16 seg.); +322 m, +8.8 m/s (12 seg.)
grid independence	293.20 t vs 293.31 t
loss identity	≤ 0.004 m/s, all 7 cases
jax partials vs FD	$\sim 10^{-10}$
US76 vs `ambiance`	$< 0.1\,\%$ to 81 km
rocket-equation consistency	optimizer's Δv_{vac} in the hand method returns 293.3 t

The re-integration drift starts at the 440 s throttle transition, where the piecewise control is least accurate; it halves from 12 to 16 segments.

Dymos documentation: SSTO Earth-launch example (Longuski Titan II)

Limitations and What Comes Next

Known limits of v0.1 and the two remaining milestones

Known limits of v0.1

- constant C_D; no J_2, winds, 6-DOF
- reserves are inputs; the return flight is not simulated
- the ship's final 40 %-throttle arc should be re-checked with grid refinement and another initial guess
- a "Falcon-9-like" single-stage test vehicle (T/W 3.75, 95 % propellant fraction) does not converge as a max-payload problem – ≈ 0 t at 79 g_0; not a useful test case

Milestones

- **by 11/19** – $C_D(M)$, J_2, grid refinement; in-space Δv module (Hohmann / Lambert, C_3, aerocapture) feeding the terminal set for the Mars, Moon and Titan missions
- **by 12/3** – co-design: engine group from $(p_c, \varepsilon, O/F)$ via c^*, C_F; stage mass model; dry masses and loads as design variables
- then – FFSC-cycle / turbopump models as OpenMDAO components in the same problem (why Dymos)

Running It

conda environment, one command per case, warm starts and deck overrides for sweeps

```
conda env create -f environment.yml && conda activate ascentopt # cyipopt must be present
python scripts/run_case.py --vehicle vehicles/starship_v3_baseline.yaml --out runs/starship \
    --h-target 275e3 --lat 25.997 --q-max 35e3 --a-max 39.2266 --segments 16 --max-iter 400 --simulate
python scripts/run_case.py ... --out runs/heavy --warm-from runs/starship/*_out/dymos_solution.db \
    --set stage0.dry_mass=250e3 --set stage1.dry_mass=120e3
python scripts/make_figures.py runs/starship --out results/figs
bash reproduce_demo.sh # baseline + six-case sweep + figures, 15-25 min
```

Stack: OpenMDAO 3.41, Dymos 1.16, pyoptsparse 2.17 + cyipopt 1.7 / Ipopt 3.14, jax 0.7 – conda-forge + PyPI.
Deck: YAML – engines (anchor pair + exit area), stages (dry, propellant, reserve, engine sets), diameter, C_D.

Outputs: `result.json` (payload, staging, losses, constraint activity, simulate check), Dymos record for warm starts and plots, time-history / path / loss / steering figures.
Repository: `Modern-Aerospace-Propulsion/Week 10 - Mission Analysis/ascentopt/` – fixed-vehicle mode now; co-design mode to follow.

Summary

From a forward simulation to a design tool

- `ascentopt` turns the ascent simulation into an optimizer: same physics, steering and throttle solved for, stages as linked phases, orbit as a constraint, payload as the objective.
- Starship Block 3 to 275 km $\times$ 26°: 293 t on an expendable trajectory with 200/100 t dry masses; losses 1833 m/s, decomposed; max-q and 4 g_0 limits both active; staging at 145 s / 60 km / 2.61 km/s.
- Sensitivities from a seven-case sweep: ship dry mass **1:1**, booster dry mass 0.16:1, 8 % booster reserve -89 t with the staging point moving on its own.
- Verified: benchmark reproduced, explicit re-integration within 211 m / 4.5 m/s, grid-independent to 0.04 %, Δv identity closed to 1 mm/s.
- Next: $C_D(M)$ and the in-space module for the project missions; then propulsion co-design in the same NLP.